Blue River have come up with robots that can identify weeds in real time using cameras and communicate the information to another piece of machinery through telematics to decide on the appropriate action that needs to be taken for dealing with them.

Once crops are harvested, impure crop shipments can be avoided through electronic security tagging. This enables transparency to a great extent. Once consumers are aware of the entire supply chain, it is easier to prevent food-borne diseases.

Livestock monitoring

The IoT can also be used to gather information about the location and health of cattle. Installing cameras around barns and pastures enables remote monitoring of animals from anywhere. Collars with radio frequency identification (RFID), which can be low, high or ultra-high frequency, GPS and biometrics can automatically identify and provide important data about the livestock, such as birth date, mother's and father's ID numbers, offspring, veterinarian records and so on in real time.

Apps like Connecterra IDA are used for getting updates about the animals. This way each animal gets special attention and a database of their history can be maintained. This also assists in easily locating and separating sick animals from the rest to prevent spreading of disease. For example, CowManager by Agis Automatisering is a cattle-tracking sensor and software solution.

For preventing animals from going to unwanted places, conventionally low-voltage shocks and sound stimuli have been used. Several organisations are building harmless and cost-effective alternative solutions for this. Vence, for instance, has created a virtual fencing wearable for cattle.

Ultrasounds can check the quality of meat that can be obtained from an animal even before reaching the market. Research is ongoing to create different kinds of foods (meat and drinks) by manipulating genes in the future.

Software and platforms for data management

The IoT sensors collect a huge amount of data that cannot be stored on a conventional database system. Cloud-based data storage and end-to-end IoT platforms ensure that data related to weather, livestock and crop get analysed using tools for predictive analytics.

Many apps available in the market can be used for real-time data collection by field teams and to send updates. Software platforms like Cropio pile data from different hardware devices used in precision agriculture on a central platform for processing and analysing, which is crucial for making effective choices for managing farm operations. Data platforms provide aggregated data to farmers at a single location as a resource. Task management tools can easily create budgets and operation strategies for production, and compare yields against standard values.

Challenges

Many farmers are reluctant to change because they cannot afford the expensive tools and machinery required, while others are not aware of the benefits. In cases where land is not owned by the farmers themselves, there are lower chances of investment. Besides, several farmers in developing and underdeveloped countries lack the needed educational qualifications. In developed countries, the average age of farmers is quite high. This impacts their decision-making and capability of learning new technologies.

In using all these technologies, an Internet connection is a must for strong network communication. In places with poor connectivity, farmers have to struggle to make them work, which impacts their motivation.

Implementing different technologies in agriculture is vital for effectively dealing with growing competition and getting better results while using fewer resources. Crop-harvesting robots, like those by Harvest CROO Robotics, can make up for the lack of enough workforce in developed nations. For staying in business, adoption of these measures is not optional but mandatory. Before making any decision on long-term investments, farmers should consult experts and fellow farmers regarding potential benefits. It is not only necessary for owners but also for employees to learn to work with these technologies. EFY